

Erratum Satellite co-existence

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1 Error

In the paper we proved that :

$$\mathbb{E}[e^{t\mathcal{I}}] = \exp(\lambda_N |\mathcal{F}_{y_0}| (\mathbb{E}[e^{tZ(\tilde{X})}] - 1)) \quad (1)$$

This formula is *incorrect* due to a mishandling of the conditional expectation, particularly in the transition from line (2) to line (3) in the proof of Proposition 2. Here is the correct derivation:

Theorem 1.

$$\begin{aligned} \mathbb{E}[e^{t\mathcal{I}}] &= \mathbb{E}[\exp(\lambda_N |\mathcal{F}_{y_0}| \mathbb{E}[e^{tZ(\tilde{X})} - 1 | Z])] \\ &\leq \mathbb{E}[\exp(\lambda_N |\mathcal{F}_{y_0}| (e^{tZ(\tilde{X})} - 1))]. \end{aligned} \quad (2)$$

Proof.

$$\mathbb{E}[e^{t\mathcal{I}}] = \mathbb{E}\left[\mathbb{E}\left[\prod_{X_k \in \phi \cap \mathcal{F}_{y_0}} e^{tZ(X_k)} \mid \phi(\mathcal{F}_{y_0}), Z\right]\right] \quad (3)$$

$$= \mathbb{E}\left[\left(\int_{\mathcal{F}_{y_0}} \frac{e^{tZ(x)}}{\Lambda(\mathcal{F}_{y_0})} d\Lambda(x)\right)^{\phi(\mathcal{F}_{y_0})}\right] \quad (4)$$

$$= \mathbb{E}\left[\mathbb{E}\left[\left(\int_{\mathcal{F}_{y_0}} \frac{e^{tZ(x)}}{\Lambda(\mathcal{F}_{y_0})} d\Lambda(x)\right)^{\phi(\mathcal{F}_{y_0})} \mid Z\right]\right] \quad (5)$$

$$= \mathbb{E}\left[\sum_{k=0}^{\infty} \left(\int_{\mathcal{F}_{y_0}} \frac{e^{tZ(x)}}{\Lambda(\mathcal{F}_{y_0})} d\Lambda(x)\right)^k \frac{\Lambda(\mathcal{F}_{y_0})^k}{k!} e^{-\Lambda(\mathcal{F}_{y_0})}\right] \quad (6)$$

$$= \mathbb{E}\left[\exp\left(-\int_{\mathcal{F}_{y_0}} (e^{tZ(x)} - 1) \tilde{\lambda}(x) dx\right)\right] \quad (7)$$

$$= \mathbb{E}\left[\exp(\lambda_N |\mathcal{F}_{y_0}| \mathbb{E}[e^{tZ(\tilde{X})} - 1 | Z])\right]. \quad (8)$$

$$\leq \mathbb{E}[\exp(\lambda_N |\mathcal{F}_{y_0}| (e^{tZ(\tilde{X})} - 1))]. \quad (9)$$

(3) to (4) use the independence among the different processes plus the property of Poisson process: knowing the number of points in a set W , these points are i.i.d distributed in W according to the law $\frac{\Lambda(\cdot)}{\Lambda(W)}$. (5) to (6) use again the independence between the processes and that $\phi(\mathcal{F}_{y_0})$ is distributed as a Poisson random variable of parameter $\Lambda(\mathcal{F}_{y_0})$. Finally (7) to (8) comes from writing the intensity measure of ϕ in the spherical coordinate. (8) to (9) is the Jensen equality in the good sense this time. $\square$

We realize that the mistake in the paper hence (1) is related with (2) as it consists of switching expectation and the exponential function. Since $x \rightarrow e^x$ is convex Jensen inequality says (2) $\geq$ (1) therefore the upper bound in the paper is false. One way to falls back on our foots is to bound the Jensen gap which is for f convex find M such that : $|\mathbb{E}[f(X)] - f(\mathbb{E}[X])| \leq M$. This will allow us to have (2) $\leq$ (1) + M and use the tools we developed before to deal with (1).

2 One way of correcting it

Before continuing, maybe it is sufficient to stop at (2) and plug the optimal t from the paper in here, but then we will fail to have a kind and gentle closed form expression which does not imply heavy computation.

A easy way to get bound on the Jensen gap is when we deal with bounded random variable. Indeed it is possible to show the following theorem :

Theorem 2. *Let X a bounded random variable. Let $f \in C^2$. Define $M = \max_{x \in \text{supp}(X)} f''(x)$. Then : $\mathbb{E}[f(X)] - f(\mathbb{E}[X]) \leq \frac{1}{2} \text{Var}(X)M$*

Proof. On $\text{Supp}(X)$, $f'' \leq M$ hence $g : \text{Supp}(X) \rightarrow \mathbb{R}, x \mapsto f(x) - \frac{1}{2}Mx^2$ is concave. Thus we have :

$$\mathbb{E}[g(X)] \leq g(\mathbb{E}[X]) \quad (10)$$

$$\Rightarrow \mathbb{E}[f(x) - \frac{1}{2}Mx^2] \leq f(\mathbb{E}[X]) - \frac{1}{2}M\mathbb{E}[X]^2 \quad (11)$$

$$\Rightarrow \mathbb{E}[f(X)] - f(\mathbb{E}[X]) \leq \frac{1}{2}\text{Var}(X)M \quad (12)$$

$\square$

With the framework of the paper $\mathbb{E}[e^{tZ(\tilde{X})} - 1|Z]$ is not bounded since $Z(x)$, which represent the interference coming from cluster at position x , can be infinite. Indeed with the model of the paper the number of transmitters accounting for down-link (resp. uplink) per cluster is distributed as Poisson random variables which is light tail but infinite support. In addition the fading $H_{b,k}$ are distributed as Gamma random variables which also have infinite support. Therefore it's very unlikely but our models deals with potential realization where the number of transmitters is really huge and where the different obstacles during the transmission behaves as huge amplifiers. Hence we can not use Theorem

2 right away. It might be possible to get a bound on the Jensen gap in this setting playing on the fact that events where $Z(x)$ blows up are really rare but it's beyond the scope of the paper. However we can change the setting and take a number of transmitters per cluster and the fadings bounded. We will loose in generality but transpose in a setting where we can use what we did before.

2.1 New setting

Replace N_x and $\tilde{N}_x$ by $N_x \wedge S_B$ and $\tilde{N}_x \wedge S_U$. Same with the $H_{U,k}$ and $H_{B,k}$ they become $H_{U,k} \wedge \tilde{S}_U$ and $H_{B,k} \wedge \tilde{S}_B$.

Hence $Z(x) \leq C := \max_{\mathcal{F}_{y_0}} \delta_U(x) S_U G_U \tilde{S}_U + \max_{\mathcal{F}_{y_0}} \delta_B(x) S_B G_B \tilde{S}_B$ and

$$\lambda_N |\mathcal{F}_{y_0}| (e^{tZ(\tilde{X})} - 1) \leq \lambda_N |\mathcal{F}_{y_0}| (e^{tC} - 1) := M$$

Hence:

Theorem 3. $\mathbb{E}[e^{t\mathcal{I}}] \leq \exp\left(\lambda_N |\mathcal{F}_{y_0}| \left(\mathbb{E}[e^{tZ(\tilde{X})}] - 1\right)\right) + \frac{1}{2}(\lambda_N |\mathcal{F}_{y_0}|)^2 \mathbb{V}ar(e^{tZ(\tilde{X})})M$

Proof. Use Theorem 2 for $X = \lambda_N |\mathcal{F}_{y_0}| (e^{tZ(\tilde{X})} - 1)$ and $f = x \rightarrow \exp(x)$ $\square$

Finally we have :

$$\mathbb{P}(\mathcal{I} \geq \tau) \leq \exp\left(-t\tau + \lambda_N |\mathcal{F}_{y_0}| \left(\mathbb{E}\left[e^{tZ(\tilde{X})}\right] - 1\right)\right) + \frac{1}{2}e^{-t\tau}(\lambda_N |\mathcal{F}_{y_0}|)^2 \mathbb{V}ar(e^{tZ(\tilde{X})})M \quad (13)$$

From here we can proceed in the same fashion than in the paper. It consists to fit $Z(\tilde{X})$ to a kind distribution for which we can compute $\mathbb{E}\left[e^{tZ(\tilde{X})}\right], \mathbb{V}ar(e^{tZ(\tilde{X})})$. Then find the optimal t for this fitted version in closed form in (13) and plug it. Note that here we do not have to be careful about the domain of convergence since everything is bounded. Once the optimal t plugged we can estimate through polynomials the quantities $\mathbb{E}\left[e^{tZ(\tilde{X})}\right], \mathbb{V}ar(e^{tZ(\tilde{X})})$ as we did in the first paper and end up with a closed form with little computations.